

Test Of Hypothesis For Large Samples And Small Samples

When performing a hypothesis test, the size of the sample (n) is crucial because it determines which statistical distribution and formula we use. The general dividing line between a large and a small sample is a sample size of $n = 30$.



Difference between Large and Small Sample

Feature	Large Sample ($n > 30$)	Small Sample ($n \leq 30$)
Sample Size	The sample size is greater than 30 ($n > 30$).	The sample size is less than or equal to 30 ($n \leq 30$).
Population Estimate	The value of the sample statistic (like the sample standard deviation, S) is a very reliable estimate of the corresponding population parameter (σ).	The sample statistic is not a reliable enough estimate of the population parameter.
Test Distribution	The Normal distribution (using the Z-test) is used for testing, thanks to the Central Limit Theorem .	Sampling distributions like the t-distribution , F -distribution, or Chi-square (χ^2) distribution are used for testing.

// Analogy: A large sample ($n > 30$) gives you enough information to assume the sample mean is distributed normally, allowing you to use the standard Z -test. A small sample ($n \leq 30$) is riskier and requires a more conservative approach using the **t-distribution**, which accounts for this greater uncertainty.



Test Of Hypothesis For Large Samples ($n > 30$)

For large samples, we primarily use the **Z-test**. This approach is justified by the **Central Limit Theorem**, which states that the distribution of sample means tends to be normal, regardless of the population's original distribution, as long as the sample size is large.

Two important applications include:

1. **Test of hypothesis about a single population mean** (μ).
2. **Test of hypothesis about the difference between two population means** ($\mu_1 - \mu_2$).

Test Of Hypothesis About Population Mean

This test is used when we want to check if the true average (μ) of a population is equal to a specific claimed value (μ_0).

1. Formulating Hypotheses:

- **Null Hypothesis** (H_0): $\mu = \mu_0$ (The population mean is the specified value).
- **Alternate Hypothesis** (H_1): $\mu \neq \mu_0$ (Two-tailed) OR $\mu > \mu_0$ (Right-tailed) OR $\mu < \mu_0$ (Left-tailed).

2. Test Statistic (Z):

The formula used to calculate the test statistic, Z , tells us how many standard errors the sample mean ($\bar{x}$) is away from the hypothesized population mean (μ):

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}$ = Sample Mean
- μ = Hypothesized Population Mean (from H_0)
- n = Sample Size
- σ = Population Standard Deviation

Important Note for Large Samples: If the true population standard deviation (σ) is unknown (which is common), we can confidently substitute it with the **Sample Standard**

Deviation (S) because $n > 30$. That is, $\sigma \approx S$.

3. Critical Values (Z_{Critical}):

The critical value is found using the standard Normal Distribution table based on the chosen level of significance (α) and the type of test (one or two-tailed).

Level of Significance (α)	Two-Tailed Test ($H_1 : \mu \neq \mu_0$)	One-Tailed Test ($H_1 : \mu > \mu_0$ or $\mu < \mu_0$)
1% ($\alpha = 0.01$)	Z = 2.58	Z = 2.33
5% ($\alpha = 0.05$)	Z = 1.96	Z = 1.645

4. Conclusion:

- Calculate the value of Z using the formula.
- Compare the absolute calculated Z -value ($|Z|$) with the appropriate critical Z -value from the table.
- If $|Z_{\text{Calculated}}| \leq |Z_{\text{Critical}}|$, **Accept H_0** .
- If $|Z_{\text{Calculated}}| > |Z_{\text{Critical}}|$, **Reject H_0** .